Introduction to cost terms and purposes

Cost terminology

Cost—a resource used to achieve a specific objective

Actual cost—a cost that has occurred

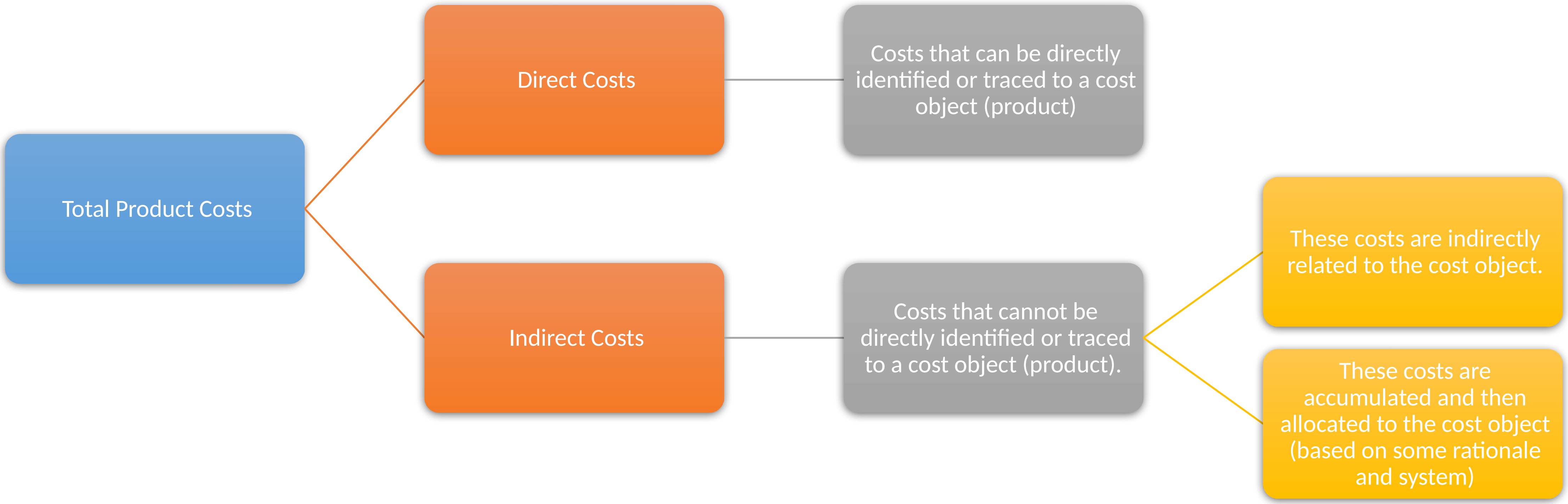
Budgeted cost—a predicted cost

Cost object—anything for which a cost measurement (resources used) is desired

Examples of cost objects at Apple Inc.

- Cost object is a Product
  - Apple iPhone
- Cost object is a Service
  - Hotline providing help to Apple store customer service agents
- Cost object is a Project
  - R&D for iPhone 15
- Cost object is a Customer
  - ABC Corporation that provides iPhone to all their salespersons
- Cost object is an Activity
  - Setting up machine for production of iPhone
- Cost object is a Department
  - Legal department

Product costs - Direct and indirect costs



Example

Direct Costs	Direct material like processor for iPhone, case, screen, etc.
	Direct labor – labor cost of employee assembling the iPhone
Indirect Cost	Factory rent
	Factory administration expenses
	Plant maintenance expenses

Other terms

Cost Tracing

Cost Accumulation

Cost Assignment

Tracing a direct relationship from a resource use to a cost object (product, service, department, etc.)

Gathering (collecting) costs (or resources used) in an organized manner (department, activity, etc.)

Allocating the accumulated costs to cost objects that have an indirect relationship to resources used.

Cost Allocation - example

Microprocessor

Direct Cost

Indirect Cost

Lease cost for factory

Cost Allocation

Direct Costs – Relatively easy to allocate

- Trace from resource to cost object
- Tracing cost of microprocessor to iPhone

Indirect costs – More challenging since these costs cannot be traced

- Must be allocated based on a rational formula
- Allocating lease cost of factory to each iPhone.

Other considerations

Costs that are direct for one object may be indirect for another object

The materiality of the cost and the expense associated with tracing it to a cost object determines whether a cost will be treated as direct or indirect

Measurement precision desired

Product and Period costs

Total Costs	Product Costs	necessary to manufacture the product	Material Labor Manufacturing Overheads
	Period Costs	non manufacturing costs expensed during the period	Selling and distribution expenses Rent on office space Marketing expense

Classifications based on cost behavior

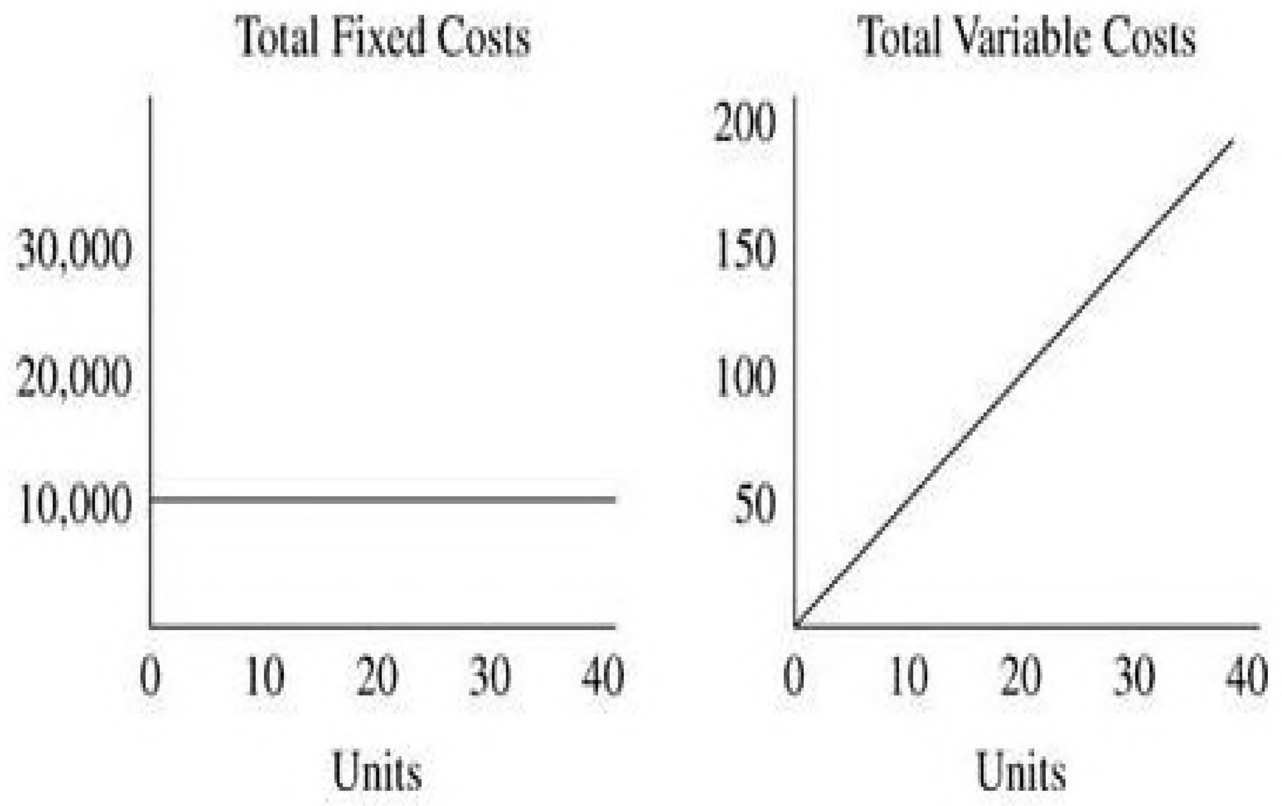
Total VARIABLE costs change in relation to output. Greater output entails greater cost.

Total cost of microprocessors – More iPhones produced – more microprocessors used – greater cost.

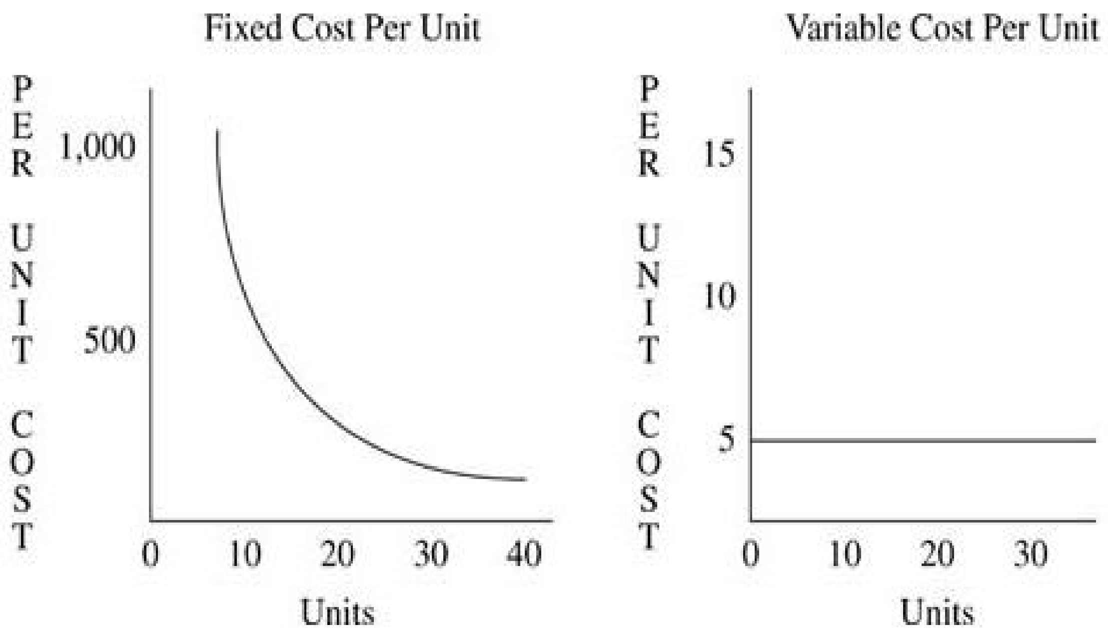
Total FIXED costs remain the same regardless of output.

Factory Rent – remains the same regardless of how many iPhones are produced

Cost graphs


Total Fixed Costs
30,000
20,000
10,000
0
10
20
30
40
Units
Total Variable Costs
200
150
100
50
0
10
20
30
40
Units

Cost behavior per unit



Fixed cost PER UNIT decreases as output increases

- Greater output (volume) leads to lower costs (efficiency)

Variable cost PER UNIT remains the same.

Some more concepts.......

Mixed Costs – contain elements of both fixed and variable costs – in the total variable cost graph think of a positive Y intercept.

Cost Driver – some activity that causes cost to increase (the reason for resource use)

Relevant range – the range of output (or time) for which the relationship between volume and costs are fixed

Example – Mixed Cost

- Annual expenses of operating an automobile – part of the cost relates to the miles driven (gasoline) and part of cost is independent of the miles driven (insurance, lease payment)

Mixed Costs

Cost Driver – In this example, ”miles driven” is the cost driver.

Example - Relevant Range

Some fixed costs may be fixed only up to a certain level of activity.

For instance, annual lease may be fixed at say $10,000 up to 12,000 miles. Anything driven more than 12,000 miles may entail additional cost.

So 0-12,000 miles is the relevant range for the fixed cost.

Multiple classifications

Direct/Variable/Product

- Cost of microprocessors for iPhone 11

Direct/Fixed/Product

- Salary of supervisor who oversees iPhone 11 production in a factory where many other iPhone models are also made

Indirect/Variable/Product

- Power cost at factory where many iPhone models are made and metered at model level.

Indirect/Fixed/Product

- Rent for the factory where many other iPhone models are made.

Commonly used cost terms – manager defined

Total Cost

- Total fixed cost + (Variable cost per unit x Number of units)

Average Cost

- Total cost ÷ Number of units produced

Marginal Cost

- The cost of making one more unit.
- Fixed costs will not change if one more unit is made, *unless* the plant is already operating at 100% capacity.